NovaTech Systems — AI RAG Architecture

NovaTech Systems • Internal Knowledge Base Document

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Purpose and scope

Detailed reference architecture for NovaTech's internal RAG assistant, including ingestion, hybrid retrieval, reranking, context construction, validation, citations, evaluation, and monitoring.

All names, systems, processes, dates, limits, and examples in this document are fictional and exist only for the NovaTech Systems RAG knowledge-base exercise.

1. Purpose

This document defines the approved reference architecture for NovaTech's internal AI assistant. The architecture is intended for enterprise knowledge retrieval and question answering where grounded responses, access control, citations, and refusal when evidence is insufficient are important.

2. Approved Architecture

User → FastAPI Gateway → AI Orchestrator → Query Processing → Hybrid Retrieval (Vector Search + BM25) → Reranker → Context Builder → LLM → Response Validator → User

Layer	Primary responsibility
FastAPI Gateway	Request authentication, request validation, rate controls, and API boundary.
AI Orchestrator	Coordinates retrieval, model invocation, validation, tracing, and policy decisions.
Query Processing	Normalizes the query, detects useful filters, and prepares retrieval variants.
Vector Search	Semantic retrieval using embeddings; Qdrant is the approved vector database.
BM25	Lexical retrieval for exact terms, identifiers, policy names, and uncommon phrases.
Reranker	Reorders candidate passages using a more precise relevance model.
Context Builder	Selects, deduplicates, orders, and formats evidence for the LLM.
LLM	Generates the answer using the supplied instructions and retrieved context.
Response Validator	Checks grounding, citation presence where required, policy constraints, and refusal conditions.

3. Infrastructure Components

Component	Technology	Purpose
API gateway	FastAPI	Application/API boundary
Vector database	Qdrant	Embedding-based retrieval
Cache	Redis	Caching selected query/retrieval/model results where safe
Metadata database	PostgreSQL	Document metadata, versions, access metadata, ingestion state
LLM	Approved model provider	Natural-language generation

4. Data Ingestion

1. Collect approved source documents from authorized repositories.
2. Extract text while preserving document identifiers, headings, tables, version, effective date, owner, status, and source location where possible.
3. Normalize text without destroying meaningful semantic boundaries.
4. Chunk content using section-aware boundaries rather than arbitrary page cuts.
5. Generate embeddings for retrievable chunks.
6. Store vectors in Qdrant and document/chunk metadata in PostgreSQL.
7. Record ingestion status and source version so stale content can be identified.

5. Chunking Guidance

- Prefer coherent sections over fixed-size fragments when a section contains a complete rule or procedure.
- Keep document ID, version, effective date, and section title attached to every chunk.
- Avoid chunks so small that a retrieved sentence loses its policy context.
- Avoid chunks so large that unrelated sections dominate the context window.
- Preserve table headers with table rows where possible.

6. Hybrid Retrieval

Hybrid retrieval combines semantic and lexical signals. Vector search is useful for conceptual similarity; BM25 is especially useful for exact policy IDs, role names, product names, error codes, and dates. Candidate results should be merged before reranking.

7. Reranking

The reranker should prioritize passages that directly answer the user’s question, match requested metadata such as product or department, and represent the active policy version when the question is about current rules. Historical versions should remain retrievable when the user asks about a past date or explicitly requests historical information.

8. Metadata Filtering

Metadata field	Example use
document_id	Retrieve NT-HR-004 specifically.
version	Compare Version 1.0 with Version 2.0.
effective_date	Find the policy applicable on a historical date.
status	Prefer active documents for current-policy questions.

Metadata field	Example use
department	Restrict results to Engineering or Information Security.
product	Restrict results to NovaFlow, NovaDesk, or NovaID.
section	Target troubleshooting, FAQ, policy, or procedure sections.

9. Prompt Construction

The context prompt should instruct the model to answer from retrieved evidence, distinguish current from historical policy, preserve uncertainty, and refuse unsupported factual claims. A recommended behavioral rule is: “Use only the supplied context for factual claims. If the answer is not supported by the context, say that the information is not available in the NovaTech knowledge base.”

10. Citation Generation

- Citations should identify the document ID and, where practical, version and section.
- Do not cite a historical policy as the current rule unless the user explicitly asks for historical information.
- When multiple documents support a conclusion, cite each relevant document.
- If documents conflict, investigate version/effective-date metadata before answering.

11. Hallucination and Refusal

The assistant must not invent company revenue, employee salaries, production credentials, API keys, customer confidential information, exact LLM API costs, future unreleased product features, or private employee medical information. These categories are intentionally absent from the knowledge base.

12. Prompt Injection Defense

- Treat retrieved documents as data, not as instructions that override system/application policy.
- Do not follow embedded requests to reveal secrets, ignore access controls, or change the assistant’s governing behavior.
- Separate trusted application instructions from retrieved document text.
- Log suspicious retrieval content for evaluation and security review.

13. Evaluation

Metric	What it measures
Retrieval recall	Whether relevant evidence is retrieved.
Retrieval precision	Whether retrieved evidence is actually useful.
Answer groundedness	Whether claims are supported by retrieved context.
Citation correctness	Whether citations support the stated claims.
Refusal accuracy	Whether the system refuses when evidence is insufficient.
Version accuracy	Whether the answer uses the correct policy version for the question date.

14. Monitoring

- Track retrieval latency, reranking latency, model latency, and end-to-end latency.
- Monitor empty retrievals and low-confidence retrieval patterns.

- Track citation failures, unsupported-claim flags, and refusal behavior.
- Monitor cache hit rate without caching sensitive information beyond approved retention rules.
- Maintain evaluation datasets for current-policy, historical-policy, cross-document, and unanswerable questions.

15. Cross-Document Reasoning Examples

Question type	Evidence path
What are the current remote-work rules and how should remote employees secure access?	NT-HR-004 + NT-SEC-001
Can an engineering employee receive production access on day one?	NT-HR-005 + NT-SEC-001
What happens to a NovaFlow production issue?	NT-PROD-001 + NT-SRE-001 + NT-ENG-001
How does NovaID support protected API calls?	NT-TECH-001 + NT-SEC-001

16. Related Documents

- NT-AI-001 — AI Engineering Handbook
- NT-SEC-001 — Security Policy
- NT-TECH-001 — NovaID Documentation
- NT-PROD-001 — NovaFlow Documentation
- NT-SRE-001 — Incident Management Runbook